

Secure Multi-Party Computation (MPC)

Enabling Private Collaboration for AI Systems

Multiple parties computing functions jointly while keeping their inputs private - a core pillar of privacy-preserving AI.



Why MPC Matters

Organizations often need to collaborate on AI models, but face significant barriers:

Privacy Constraints

Data cannot be shared due to confidentiality requirements

Regulatory Restrictions

Legal frameworks like GDPR, HIPAA limit data sharing

Competitive Concerns

Proprietary data represents business advantage



Example: Banks detecting fraud patterns across institutions without revealing sensitive transaction data

The Core Concept of MPC

01

Split Data into Shares

Sensitive information is mathematically divided into separate pieces

02

Distributed Computation

Each party performs calculations only on their share of the data

03

Secure Combination

Results are combined to reveal only the final outcome

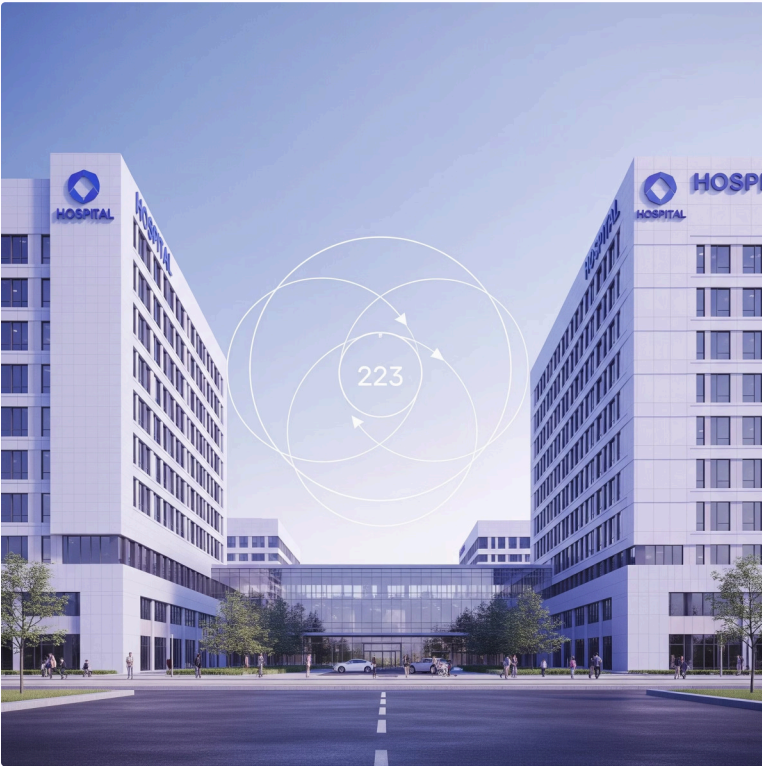
04

Privacy Preserved

No party learns anyone else's private data inputs

This cryptographic approach enables [collaborative intelligence](#) while maintaining [data confidentiality](#).

Classic MPC Example: The Hospital Dilemma



The Challenge:

Two hospitals want to discover how many patients they have in common without revealing patient identities.

The MPC Solution:

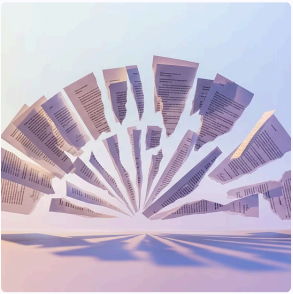
Using secure set intersection protocols, hospitals compute **only the size** of their patient overlap.



Output: "37 patients in common"

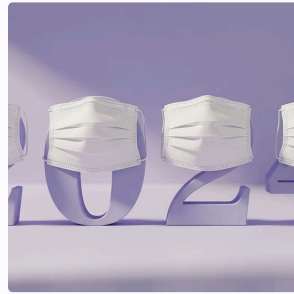
Not revealed: Which specific patients or any personal data

Key Techniques Used in MPC



Secret Sharing

Split inputs into random shares where individual pieces reveal nothing about the original



Additive Masking

Values hidden by adding random numbers that cancel out during computation



Garbled Circuits

Encode computations into encrypted truth tables that reveal only final outputs



Homomorphic Encryption

Perform calculations directly on encrypted values without decryption

Additive Secret Sharing Example

For a sensitive value $x = 42$:



Split Into Shares

Party A receives: 15

Party B receives: 27

$$(15 + 27 = 42)$$



Individual Security

Neither party knows the true value alone

Each share appears random



Secure Computation

Parties perform calculations on shares

Results combine to reveal answer



This simple concept extends to complex ML calculations through specialized protocols

MPC in Machine Learning

Secure Training

- Multiple organizations contribute encrypted gradients
- Model weights updated without exposing data
- Privacy-preserving backpropagation

Private Inference

- Encrypted queries processed across private datasets
- Model predictions without revealing inputs
- Split model architecture across parties

Complementary Technologies

- Combines with [Federated Learning](#)
- Enhances [Differential Privacy](#)
- Used in regulated industries

MPC provides cryptographic guarantees for privacy in collaborative AI systems across **finance, healthcare, and government** applications.

Real Example: Secure Linear Regression

The Business Problem:

Multiple banks want to build a joint credit scoring model without exposing customer financial data.

MPC Implementation:

1. Each bank encrypts customer features (income, assets, history)
2. Secure protocol computes regression coefficients
3. Only final model parameters are revealed
4. Raw customer data never leaves bank systems



- ❏ Cryptographic guarantees ensure each bank's competitive data remains private while creating a more robust joint model.

Leading MPC Tools & Frameworks



CrypTen

Facebook's PyTorch-based library for privacy-preserving machine learning, optimized for research and enterprise deployment



TF Encrypted

Integrates MPC protocols with TensorFlow, allowing secure computation directly within standard TensorFlow workflows



MP-SPDZ

General-purpose MPC system supporting multiple protocols with different security/performance tradeoffs



PySyft

OpenMined's framework combining MPC with federated learning and differential privacy for comprehensive privacy

These frameworks make implementing secure computation increasingly accessible to ML engineers.

Key Advantages of MPC

Cryptographically Sound Privacy

Based on mathematical guarantees rather than trust or policies

Enables Previously Impossible Collaboration

Organizations can safely work together while protecting intellectual property and sensitive data

Broad Data Compatibility

Works with structured tabular data and unstructured content (images, text) through specialized protocols

Regulatory Compliance

Helps meet GDPR, HIPAA, and other privacy requirements by keeping data encrypted



MPC provides [provable security guarantees](#) based on cryptographic hardness assumptions

Current Limitations of MPC



These challenges drive ongoing research in cryptographic optimization and hardware acceleration

→ Performance Overhead

10-1000× slower than plaintext computation depending on protocol, with high communication costs

→ Coordination Requirements

All parties must be online and synchronized during computation phases

→ Implementation Complexity

Implementing large neural networks requires specialized expertise and careful protocol selection

→ Scaling Challenges

Most protocols scale poorly with participant count, making million-user deployments difficult

MPC in Production: Real-World Applications



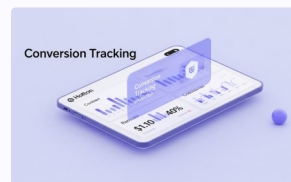
Financial Services

Cross-bank fraud detection and risk modeling without exposing customer transactions



Healthcare

Multi-hospital diagnostic models and rare disease research preserving patient confidentiality



Digital Marketing

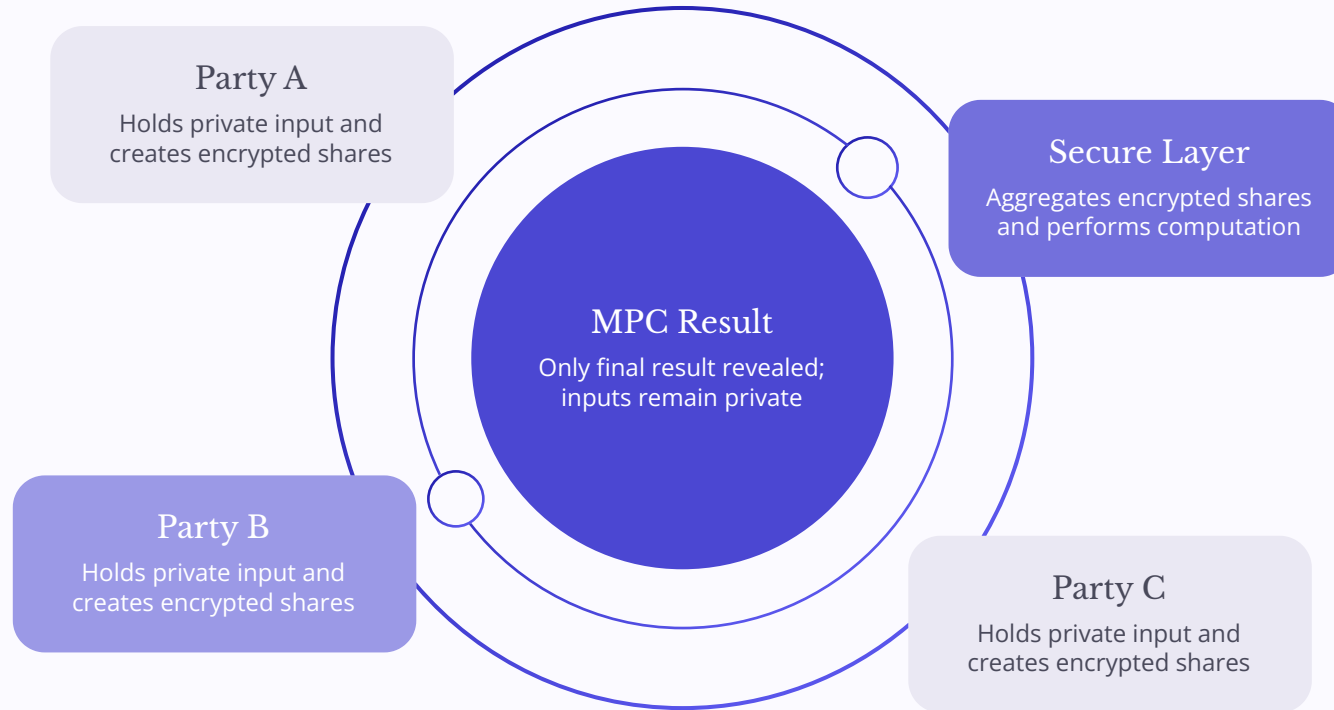
Privacy-preserving ad conversion measurement across platforms without tracking individuals



Government

Joint intelligence and defense analytics while protecting classified information sources

How MPC Works: The Flow of Secure Computation



The cryptographic protocols ensure computation correctness while mathematically preventing access to the original inputs from any single party.

Future Directions in MPC Research



Hybrid Privacy Stacks

Combining MPC + Federated Learning + Differential Privacy for balanced systems



Hardware Acceleration

Specialized chips and trusted execution environments to dramatically improve performance



Zero-Knowledge Proofs

Verifiable computation ensuring parties follow protocols without revealing data



Production Scaling

Moving from research prototypes to enterprise-ready, high-performance systems



Current research focuses on reducing the significant performance gap between secure and plaintext computation

Summary: The Promise of MPC

Core Concept

Joint computation without data sharing, based on cryptographic guarantees

Key Techniques

Secret sharing, homomorphic encryption, garbled circuits, and secure multi-party protocols

Primary Applications

Finance, healthcare, marketing, and government AI systems requiring privacy

While computational overhead remains a challenge, MPC works best when paired with complementary techniques like federated learning and differential privacy.

 **Next Steps:** Explore how MPC fits into your organization's privacy-preserving AI strategy, particularly for cross-organizational collaborations where data sharing is restricted.